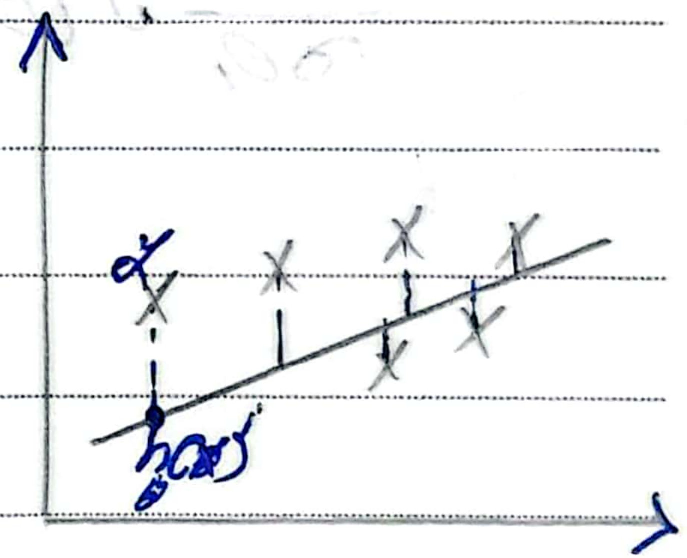
* Cost Function

$$\frac{1}{2m}\sum_{i=1}^{m}\left(h_\theta(x^{(i)})-y^{(i)}\right)^2$$

→ Square Error Function



KAYAN

* R^2 square :

$$R^2 = 1 - \frac{\sum (y_i - \overset{\xrightarrow{h_\theta(x)}}{\hat{y}_i})^2}{\sum (y_i - \bar{y})^2} = 1 - \frac{\text{Low}}{\text{High}}$$

* Location Price $R^2 = 85\%$
* Bedrooms Location Price $R^2 = 90\%$ $P=2$
* Gender Bedrooms Location Price $R^2 = 91\%$ $P=3$

→ note: at third case we have not important feature [Gender] and R^2 still increase to solve this problem we use adjusted R^2

* Adjusted R^2

$$= 1 - \boxed{\frac{(1-R^2)(N-1)}{N-P-1}}$$

N; number of data Point

EX, $P=2$ $R^2 = 90\%$ R^2 adjusted $= 85\%$

$P=3$ $R^2 = 91\%$ R^2 adjusted $= 80\%$

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2$$

$$P = \lambda \sum_{i=1}^{m} w_i^2 \longrightarrow \text{Pelant } \alpha$$

$$Cost = \frac{1}{2m}\sum_{i=1}^{m}(\hat{y}_i - y_i)^2 + \lambda \sum_{i=1}^{m} w_i^2$$

Performance matrix

if you have:

y	$\hat{y}$
0	1
1	1
0	0
1	1
1	1
0	1
1	0

Actual

Perd	1	0
1	3	2
0	1	1

Actual

Perd	1	0
1	TP	FP
0	FN	TN

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN}$$

$$= \frac{4}{7} = 57\%$$